Userspace ALSA

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- The main way to interact with ALSA devices is to use alsa-lib.
- `https://github.com/alsa-project/alsa-lib`
- It provides mainly access to the devices but also goes further and allows handling audio in userspace.
- The library itself is actually named `libasound`
- The include file is `alsa/asoundlib.h`

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- ```
  int snd_pcm_open(snd_pcm_t ** pcmp, const char * name, snd_pcm_stream_t stream,
  int mode)
  ```
- `name` is the name of the PCM to be opened.
- `stream` can be either `SND_PCM_STREAM_PLAYBACK` or

  ```
  SND_PCM_STREAM_CAPTURE
  ```
- `mode` can be a combination of `SND_PCM_NONBLOCK` and

  ```
  SND_PCM_ASYNC
  ```
- ```
  int snd_pcm_close(snd_pcm_t *pcm)
  ```
- Closes the PCM.

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- This can be specified as a hardware device. The three arguments (in order: CARD,DEV,SUBDEV) specify card number or identifier, device number and subdevice number (-1 means any). For example: `hw:0` or `hw:1,0` . Instead of the index, the card name can be used:

```
hw:STM32MP15DK,0
```

- Or through the `plug` plugin: `plug:mypcmdef` ,

```
plug:hw:0,0
```

.

- The list of available names can be generated using

```
snd_card_next
```

to iterate over all the physical cards. See `device_list` in `aplay` .

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- The next step is to handle the PCM stream parameters
- ```
  snd_pcm_hw_params_t *hw_params; int snd_pcm_hw_params_malloc(snd_pcm_hw_params_t ** ptr)
  int snd_pcm_hw_params_any(snd_pcm_t * pcm, snd_pcm_hw_params_t * params)
  ```
- This will allocate a `snd_pcm_hw_params_t` and fill it with the range of parameters supported by `pcm` .
- ```
  int snd_pcm_hw_params_set_access(snd_pcm_t *pcm, snd_pcm_hw_params_t *params,
                                   snd_pcm_access_t _access)
  ```
- This set the proper access type: interleaved or non-interleaved, mmap or not.
- ```
  int snd_pcm_hw_params_set_format(snd_pcm_t *pcm, snd_pcm_hw_params_t *params,
                                   snd_pcm_format_t val)
  ```
  ```

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- This set the format, using a `SND_PCM_FORMAT_` macro.

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- ```
  int snd_pcm_hw_params_set_channels(snd_pcm_t *pcm, snd_pcm_hw_params_t *params,
                                     unsigned int val)
  ```

- Sets the number of channels.

- ```
  int snd_pcm_hw_params_set_rate_near(snd_pcm_t *pcm, snd_pcm_hw_params_t *params,
                                      unsigned int *val, int *dir)
  ```

- Sets the sample rate, setting `dir` to 0 will require the exact rate.

- ```
  int snd_pcm_hw_params_set_periods(snd_pcm_t *pcm, snd_pcm_hw_params_t *params,
                                    unsigned int val, int dir)
  int snd_pcm_hw_params_set_period_size(snd_pcm_t *pcm, snd_pcm_hw_params_t
  *params,
                                        snd_pcm_uframes_t val, int dir)
  int snd_pcm_hw_params_set_buffer_size(snd_pcm_t *pcm, snd_pcm_hw_params_t
  ```
  ```

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```
*params,
                                                snd_pcm_uframes_t val)
```

- Sets the number of periods and the period size in the buffer or the buffer size.

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- ```
  int snd_pcm_hw_params(snd_pcm_t * pcm, snd_pcm_hw_params_t * params)
  ```
- Installs the parameters and calls `snd_pcm_prepare` on the stream.
- ```
  void snd_pcm_hw_params_free(snd_pcm_hw_params_t * obj)
  ```
- Frees the allocated `snd_pcm_hw_params_t` .
- ```
  int snd_pcm_prepare(snd_pcm_t * pcm)
  ```
- Prepares the stream.
- ```
  int snd_pcm_wait(snd_pcm_t * pcm, int timeout)
  ```
- Waits for the PCM to be ready.

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```
snd_pcm_sframes_t snd_pcm_writei(snd_pcm_t *pcm, const void *buffer,
snd_pcm_uframes_t size)
snd_pcm_sframes_t snd_pcm_readi(snd_pcm_t *pcm, void *buffer, snd_pcm_uframes_t
size)
snd_pcm_sframes_t snd_pcm_writen(snd_pcm_t *pcm, void **bufs, snd_pcm_uframes_t
size)
snd_pcm_sframes_t snd_pcm_readn(snd_pcm_t *pcm, void **bufs, snd_pcm_uframes_t
size)
```

- Write or read from an interleaved or non-interleaved buffer.

```
int snd_pcm_mmap_begin(snd_pcm_t *pcm, const snd_pcm_channel_area_t **areas,
                       snd_pcm_uframes_t *offset, snd_pcm_uframes_t *frames)
snd_pcm_sframes_t snd_pcm_mmap_commit(snd_pcm_t *pcm, snd_pcm_uframes_t offset,
                                      snd_pcm_uframes_t frames)
snd_pcm_sframes_t snd_pcm_mmap_writei(snd_pcm_t *pcm, const void *buffer,
                                      snd_pcm_uframes_t size)
snd_pcm_sframes_t snd_pcm_mmap_readi(snd_pcm_t *pcm, void *buffer,
```

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```
snd_pcm_uframes_t size)
snd_pcm_sframes_t snd_pcm_mmap_writen(snd_pcm_t *pcm, void **bufs,
snd_pcm_uframes_t size)
snd_pcm_sframes_t snd_pcm_mmap_readn(snd_pcm_t *pcm, void **bufs,
snd_pcm_uframes_t size)
```

- Write or read from an interleaved or non-interleaved mmap buffer.

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- It is possible to set controls programatically.
- ```
  snd_ctl_t *handle; int snd_ctl_open (snd_ctl_t **ctl, const char *name, int mode)
  ```
- Opens the sound card to be controlled.
- ```
  snd_ctl_elem_id_t *id;
  #define snd_ctl_elem_id_alloca(ptr)
  snd_ctl_elem_value_t *value;
  #define snd_ctl_elem_value_alloca(ptr)
  ```
- Allocate a `snd_ctl_elem_id_t` , referring to a particular control and a `snd_ctl_elem_value_t` to be set for this control.

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```
void snd_ctl_elem_id_set_interface(snd_ctl_elem_id_t *obj, snd_ctl_elem_iface_t
val)
void snd_ctl_elem_id_set_name(snd_ctl_elem_id_t *obj, const char *val)
```

- Set the interface and name of the control to be set.

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- A lookup is needed to fill the `snd_ctl_elem_id_t` completely

```
int lookup_id(snd_ctl_elem_id_t *id, snd_ctl_t *handle)
{
    int err;
    snd_ctl_elem_info_t *info;
    snd_ctl_elem_info_alloca(&info);

    snd_ctl_elem_info_set_id(info, id);
    if ((err = snd_ctl_elem_info(handle, info)) < 0) {
        return err;
    }
    snd_ctl_elem_info_get_id(info, id);

    return 0;
}
```

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```
void snd_ctl_elem_value_set_id(snd_ctl_elem_value_t *obj, const
snd_ctl_elem_id_t *ptr)
```

- Links the value with the control id.

```
void snd_ctl_elem_value_set_boolean(snd_ctl_elem_value_t *obj, unsigned int idx,
long val)
void snd_ctl_elem_value_set_integer(snd_ctl_elem_value_t *obj, unsigned int idx,
long val)
void snd_ctl_elem_value_set_integer64(snd_ctl_elem_value_t *obj, unsigned int
idx,
                                      long long val)
void snd_ctl_elem_value_set_enumerated(snd_ctl_elem_value_t *obj, unsigned int
idx,
                                       unsigned int val)
void snd_ctl_elem_value_set_byte(snd_ctl_elem_value_t *obj, unsigned int idx,
                                 unsigned char val)
void snd_ctl_elem_set_bytes(snd_ctl_elem_value_t *obj, void *data, size_t size)
```

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- Set the value in `snd_ctl_elem_value_t` .
-
```
int snd_ctl_elem_write(snd_ctl_t *ctl, snd_ctl_elem_value_t *data)
```
- Actually set the control.

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- UCM: The ALSA Use Case Configuration: https://www.alsa-project.org/alsa-doc/alsa-lib/group__ucm__conf.html
- ALSA topology: https://www.alsa-project.org/wiki/ALSA_topology

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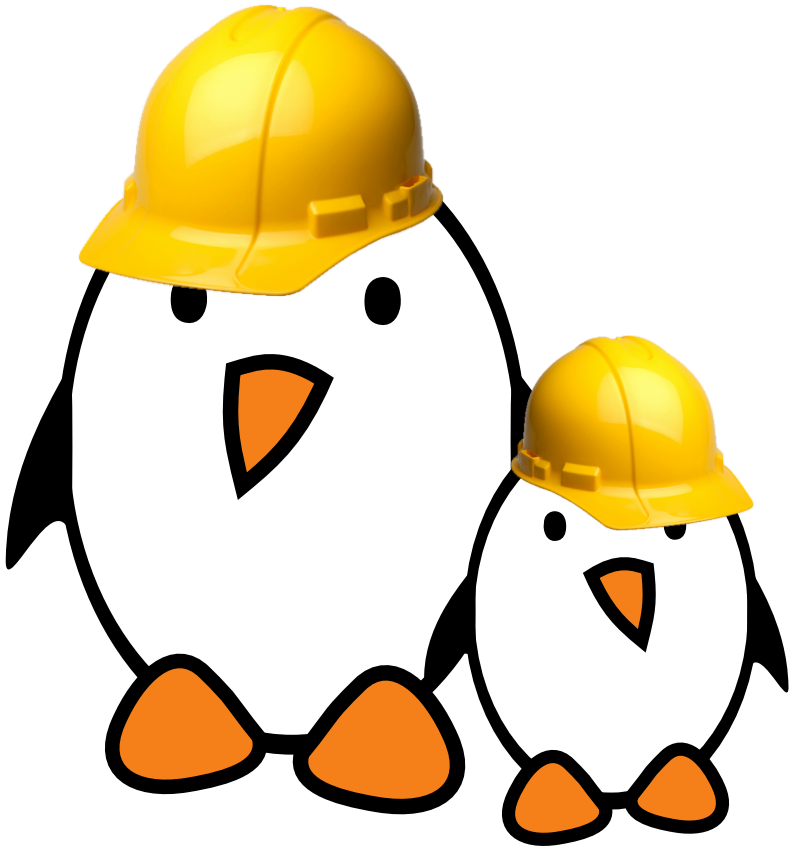

Using `alsa-lib` tools to:

- Reorder channels
- Split channels
- Resample
- Mix samples
- Apply effects

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